

## Session 6

### Aim

To study process states, process termination, zombie and orphan processes, and non-preemptive CPU scheduling.

```
teja@LAPTOP-FCGPH110:~$ ps
PID TTY          TIME CMD
134 pts/0      00:00:00 bash
453 pts/0      00:00:00 ps

teja@LAPTOP-FCGPH110:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root         1        0  ? 01:10 ?        00:00:01 /sbin/init
root         2        1  0 01:10 hvcd 00:00:00 /init
root         6        2  0 01:10 hvcd 00:00:00 plan9 --control-socket ? --log-level 4 --server-fd 8 --pipe-fd 18 --log-truncate
root        61        1  1 01:10 ?        00:00:00 /usr/lib/systemd/systemd-journald
root       187        1  1 01:10 ?        00:00:00 /usr/lib/systemd/systemd-udev
systemd+   126        1  1 01:10 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+   127        1  1 01:10 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
root       187        1  0 01:10 ?        00:00:00 /usr/sbin/cron -f -P
message+   188        1  0 01:10 ?        00:00:00 dbus-daemon --system --address=systemd: --nofork --nopidfile --system-activation --syslog-only
root       198        1  1 01:10 ?        00:00:00 /usr/lib/systemd/systemd-logind
root       281        1  1 01:10 ?        00:00:00 /usr/libexec/ssl-proc-service
root       218        1  0 01:10 tty1    00:00:00 /sbin/agetty -o -p -- \u --ncclear - linux
syslog     228        1  1 01:10 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root       288        1  2 01:10 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root       331        2  0 01:11 ?        00:00:00 /init
root       332       331  0 01:11 ?        00:00:00 /init
teja       334       332  0 01:11 pts/0    00:00:00 -bash
root       335        2  0 01:11 pts/1    00:00:00 /bin/login -f
teja       384        1  1 01:11 ?        00:00:00 /usr/lib/systemd/systemd --user
teja       385       384  0 01:11 ?        00:00:00 (sd-pam)
teja       411       335  0 01:11 pts/1    00:00:00 -bash
teja       454       334  0 01:11 pts/0    00:00:00 ps -ef

teja@LAPTOP-FCGPH110:~$ top
top - 01:11:41 up 0 min, 1 user, load average: 0.01, 0.10, 0.06
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7754.6 total, 6334.8 free, 503.7 used, 1567.8 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7256.9 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    time+  command
    1 root        20   0   21888 13172 9788 S   0.0   0.2   0:01.60 systemd
     2 root        20   0   3188  2212 2072 S   0.0   0.0   0:00.03 init-systemd(ub
    6 root        20   0   3196  2124 2088 S   0.0   0.0   0:00.00 init
   61 root       19  -1  42188 16128 15868 S   0.0   0.2   0:00.29 systemd-journal
  187 root        20   0  25152  6536 5168 S   0.0   0.1   0:00.16 systemd-udev
  126 systemd+   20   0  21304 11196 11068 S   0.0   0.2   0:00.25 systemd-resolve
  127 systemd+   20   0  91836  7916  6952 S   0.0   0.1   0:00.13 systemd-timesync
  187 root        20   0   4244  2688 2472 S   0.0   0.0   0:00.01 cron
  188 message+    20   0   9600  5000 4748 S   0.0   0.1   0:00.12 dbus-daemon
  198 root        20   0  17988  8844  7884 S   0.0   0.1   0:00.22 systemd-logind
  281 root        20   0  1756372 13688 11400 S   0.0   0.2   0:00.25 ssl-proc-service
  218 root        20   0   3124  1972  1836 S   0.0   0.0   0:00.01 agetty
  229 syslog     20   0  222516  5072  4580 S   0.0   0.1   0:00.19 rsyslogd
  239 root        20   0  107036 23172 13716 S   0.0   0.3   0:00.43 unattended-upgr
  331 root        20   0   3104  1116   988 S   0.0   0.0   0:00.00 SessionLeader
  332 root        20   0   3288  1252  1188 S   0.0   0.0   0:00.01 ReLay(334)
  334 teja        20   0   6876  5388 3668 S   0.0   0.1   0:00.07 bash
  335 root        20   0   6784  4684 3820 S   0.0   0.1   0:00.03 login
  388 teja        20   0  20328 11564  9408 S   0.0   0.1   0:00.25 systemd
  385 teja        20   0  21168  3572  1848 S   0.0   0.0   0:00.00 (sd-pam)
  411 teja        20   0   6888  5376 3788 S   0.0   0.1   0:00.07 bash
  454 teja        20   0   9388  5692 3588 R   0.0   0.1   0:00.04 top
```

```
top - 01:32:53 up 1 min, 1 user, load average: 0.26, 0.16, 0.05
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7754.6 total, 6388.4 free,  530.1 used, 1007.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7224.6 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    time+  command
    1 root        20   0   21888 13172 9788 S   0.0   0.2   0:01.68 systemd
     2 root        20   0   3188  2212 2072 S   0.0   0.0   0:00.03 init-systemd(ub
    6 root        20   0   3196  2124 2088 S   0.0   0.0   0:00.00 init
   61 root       19  -1  42188 16128 15868 S   0.0   0.2   0:00.29 systemd-journal
  187 root        20   0  25152  6536 5168 S   0.0   0.1   0:00.16 systemd-udev
  126 systemd+   20   0  21304 11196 11068 S   0.0   0.2   0:00.25 systemd-resolve
  127 systemd+   20   0  91836  7916  6952 S   0.0   0.1   0:00.13 systemd-timesync
  187 root        20   0   4244  2688 2472 S   0.0   0.0   0:00.01 cron
  188 message+    20   0   9600  5000 4748 S   0.0   0.1   0:00.12 dbus-daemon
  198 root        20   0  17988  8844  7884 S   0.0   0.1   0:00.22 systemd-logind
  281 root        20   0  1756372 13688 11400 S   0.0   0.2   0:00.25 ssl-proc-service
  218 root        20   0   3124  1972  1836 S   0.0   0.0   0:00.01 agetty
  229 syslog     20   0  222516  5072  4580 S   0.0   0.1   0:00.19 rsyslogd
  239 root        20   0  107036 23172 13716 S   0.0   0.3   0:00.43 unattended-upgr
  331 root        20   0   3104  1116   988 S   0.0   0.0   0:00.00 SessionLeader
  332 root        20   0   3288  1252  1188 S   0.0   0.0   0:00.01 ReLay(334)
  334 teja        20   0   6876  5388 3668 S   0.0   0.1   0:00.07 bash
  335 root        20   0   6784  4684 3820 S   0.0   0.1   0:00.03 login
  388 teja        20   0  20328 11564  9408 S   0.0   0.1   0:00.25 systemd
  385 teja        20   0  21168  3572  1848 S   0.0   0.0   0:00.00 (sd-pam)
  411 teja        20   0   6888  5376 3788 S   0.0   0.1   0:00.07 bash
  454 teja        20   0   9388  5692 3588 R   0.0   0.1   0:00.04 top
```

```
GNU nano 2.2 zombie.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();
    if (pid > 0)
    {
        printf("Parent process is running\n");
        sleep(20);
    }
    else if (pid == 0)
    {
        printf("Child process completed\n");
        exit(0);
    }
    else
    {
        printf("Fork failed\n");
    }
    return 0;
}
```

```
teja@LAPTOP-FGDPB110:~$ sleep 20
teja@LAPTOP-FGDPB110:~$ sleep 5
teja@LAPTOP-FGDPB110:~$ sleep 100
^C
teja@LAPTOP-FGDPB110:~$ sleep 100 &
[1] 485
teja@LAPTOP-FGDPB110:~$ jobs
[1]+  Running                  sleep 100 &
teja@LAPTOP-FGDPB110:~$ nano zombie.c
teja@LAPTOP-FGDPB110:~$ gcc zombie.c -o zombie
[1]+  Done                     sleep 100
teja@LAPTOP-FGDPB110:~$ ./zombie &
[1] 498
teja@LAPTOP-FGDPB110:~$ Child process completed
Parent process is running
ps -el | grep Z
F S  UID      PID     PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
1 Z   1000      499      498  0  80   0  -    0 -      pts/0        00:00:00 zombie
teja@LAPTOP-FGDPB110:~$
```

## Short theory

A zombie process is a child process that has completed its work, but its parent has not collected its exit status. It remains temporarily in the process table.

```
GNU nano 2.9.2 orphan.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();
    if (pid > 0)
    {
        printf("Parent process completed\n");
        exit(0);
    }
    else if (pid == 0)
    {
        sleep(5);
        printf("Child process is running\n");
        printf("New parent ID: %d\n", getppid());
    }
    else
    {
        printf("Fork failed\n");
    }
    return 0;
}
```

```
teja@LAPTOP-FGDPB110:~$ gcc zombie.c -o zombie
[1]+  Done                  sleep 100
teja@LAPTOP-FGDPB110:~$ ./zombie &
[1] 498
teja@LAPTOP-FGDPB110:~$ Child process completed
Parent process is running
ps -el | grep Z
F S  UID      PID     PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
1 Z  1000      499      498  0  80   0 -    0 -    pts/0        00:00:00 zombie
teja@LAPTOP-FGDPB110:~$ nano orphan.c
[1]+  Done                  ./zombie
teja@LAPTOP-FGDPB110:~$ gcc orphan.c -o orphan
teja@LAPTOP-FGDPB110:~$ ./orphan
Parent process completed
teja@LAPTOP-FGDPB110:~$ Child process is running
New parent ID: 332
^[[2~
```

## Short theory

An orphan process is a child process whose parent terminates before the child. The operating system assigns the child to another system process.

```

Parent process completed
teja@LAPTOP-FGDPB110:~$ Child process is running
New parent ID: 332
ps -eo pid,comm,ni,pri
  PID COMMAND          NI PRI
    1 systemd           0  19
    2 init-systemd(Ub   0  19
    6 init              0  19
   61 systemd-journal -1  20
  107 systemd-udevd     0  19
  126 systemd-resolve   0  19
  127 systemd-timesyn   0  19
  187 cron              0  19
  188 dbus-daemon        0  19
  198 systemd-logind    0  19
  218 agetty            0  19
  220 rsyslogd           0  19
  230 unattended-upgr    0  19
  331 SessionLeader      0  19
  332 Relay(334)         0  19
  334 bash              0  19
  335 login             0  19
  384 systemd            0  19
  385 (sd-pam)           0  19
  411 bash              0  19
  541 ps                0  19
teja@LAPTOP-FGDPB110:~$ nice -n 10 sleep 30 &
[1] 542
teja@LAPTOP-FGDPB110:~$ nice -n 10 sleep 30 &
[2] 543
teja@LAPTOP-FGDPB110:~$ ps -eo pid,comm,ni,pri | grep sleep
  542 sleep          10   9
  543 sleep          10   9
teja@LAPTOP-FGDPB110:~$ |

```

### Session 6 short theory

In this session, we studied process states and process termination. We observed waiting processes using sleep, normal and abnormal termination, zombie processes and orphan processes. We also observed process priority. CPU scheduling helps the operating system